13

Packaging Machines

As stated earlier in this text, the area of packaging machines is quite large. The consortium of machine producers is called the Packaging Machinery Manufacturers Institute (PMMI). The PMMI has many tasks, including organizing the largest tradeshows each year (Pack Expo) in the United States dedicated to packaging. It also has an education division, which was run for many years by the late Glenn Davis. Glenn wrote many instructional books for PMMI member companies, some for training technicians, and others summarizing the types and range of the various packaging machines (Davis, 1997). His vision lives on at PMMI, and parts of this text are inspired from his writing.

The PMMI tradeshow website (www.packexpo.com) offers a great listing of the types of packaging machines. Table 13.1 shows a high-level view of these types of machines, while the Appendix lists the complete list of over 350 types of machines. This chapter will deal with some of the most common packaging machines as an introduction. The reader is encouraged to read Glenn's work, and to view the thousands of machines in the Pack Expo website underneath the 350 product category listing.

13.1. LIQUID FILLING MACHINES

Most of us use bottles filled with liquids every day of our lives. The two-liter bottle shown in Fig. 13.1 needs to be filled with automation to keep profits high.

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TABLE 13.1 Pack Expo Packaging Machine Categories

Packaging Machinery
Adhesive/Glue
Applicators
Cartoning
Machines
Case/Tray Erecting
Machines
Case/Tray Handling
Machines
Case/Tray Loading/Unloading
Machines
Case/Tray Sealing
Machines
Case/Tray Wrapping
Machines
Cleaning
Machines
Closing
Machines
Container & Component Handling
Machines Conveyors
Cooling, Warming & Drying
Machines
Decorating
Machines
Fill & Seal
Machines
Filling
Machines
Form, Fill & Seal
Machines
Group Packaging Form, Fill & Seal
Machines
Inspecting
Machines
Labeling
Machines
Marking
Machines
Packaging Support/Specialty

(*continued*)

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Table 13.1 *Continued*

Packaging Machinery
Equipment
Pallet Forming, Dismantling & Securing
Machines
Pasteurizing
Machines
Rebuilt/Used
Equipment
Shrinking
Equipment
Sterilizing
Machines
Strapping
Machines
Wrapping
Machines

There are several ways to fill a bottle, the two biggest being volumetric and net weight. Figures 13.2 and 13.3 show a piston that is used to transfer a fixed volume into the cylinder above the empty bottle. Valves are used to allow the proper flow at the upstroke and downstroke. If the liquid density is to vary at all, this fact is ignored with this method.

For net weight filling, the empty bottle is placed on a scale (Fig. 13.4). The tank has a piston that opens a valve just above the bottle mouth. The scale is used as sensory feedback to guarantee the correct weight, even if the filled bottle looks a little under filled due to product density changes on the heavy side.



FIG. 13.1 Liquid-filled bottle

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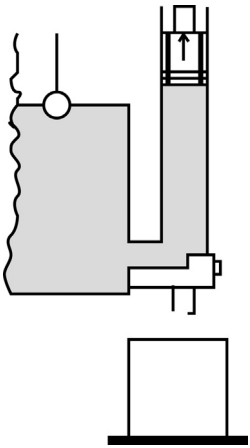


FIG. 13.2 Volumetric filling: piston up

All bottles need to be capped, some by threaded caps that get placed via a cap chute (Fig. 13.5), and are then led to rollers that will spin the cap onto the bottle threads until tight. Other caps are themselves sealed through a heat shrinking process (Fig. 13.6).

13.2. CARTONING AND BOXES

Many products (nonliquid) get placed in a cardboard box or carton. Either name can be used in practice, but the packaging industry usually states that the individ-

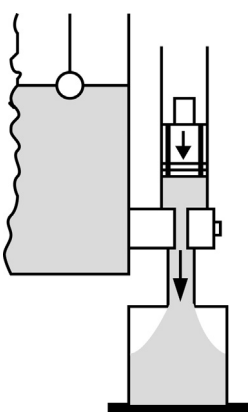


FIG. 13.3 Volumetric filling: piston down

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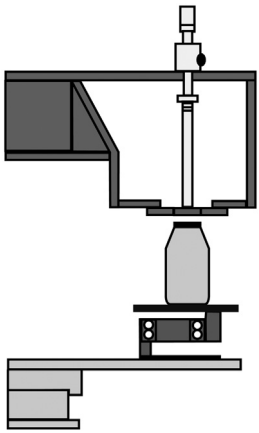


FIG. 13.4 Net weight filling: cutaway view

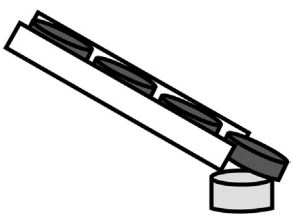


FIG. 13.5 Cap chute

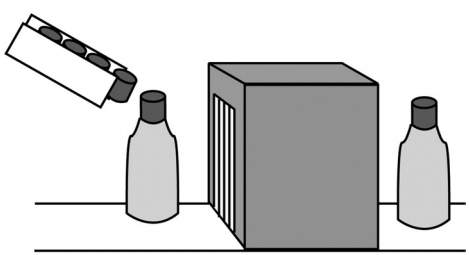


FIG. 13.6 Heat shrink capping

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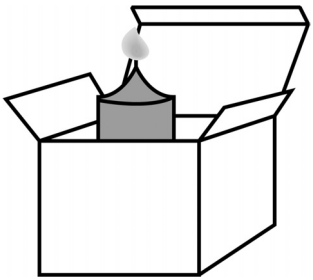


FIG. 13.7 Candle in a box: do not light before packaging!

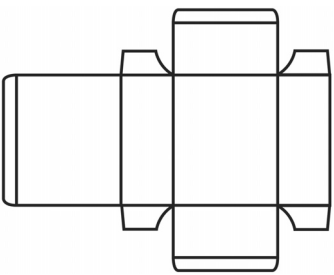


FIG. 13.8 Box in folded form for shipping

ual item is placed into a carton, and a collection of cartons placed in a box or case. When an individual item is to be cartoned (Fig. 13.7), the box first needs to be constructed or erected from a stack of folded carton blanks. These blanks come from the printer folded (Fig. 13.8) and stacked to take up less shipping space.

Carton opening (Fig. 13.9) is usually one of the first steps required in a cartoning machine (Fig. 13.10). These machines usually run with a continuous conveyor belt, and are highly synchronous in design.

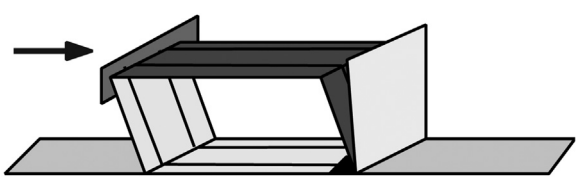


FIG. 13.9 Carton opener

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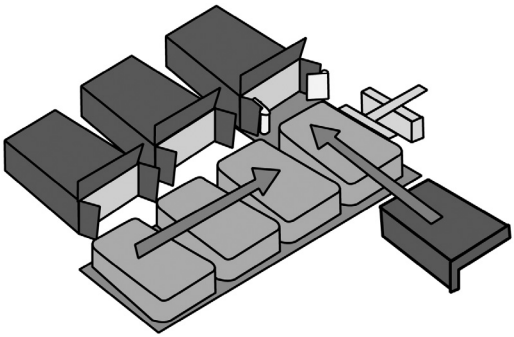


FIG. 13.10 One-shot filler cartoner

Whether it is to be used as an inside liner for a powdered product that ultimately gets cartoned (instant pudding for example), or it is to produce pouches for sale without a cartoner, the Form Fill and Seal (FFS) machine is fun to watch. It takes a roll of wrap and folds it into a continuously moving funnel. The two edges are sealed to make a tube, and then the correct amount of powdered product is dropped in. Then with careful timing, sealing fingers create a seal and cut one pouch from the next. The machine in Fig. 13.11 is shown creating an inner pouch and dropping it into a carton, all at remarkable speeds.

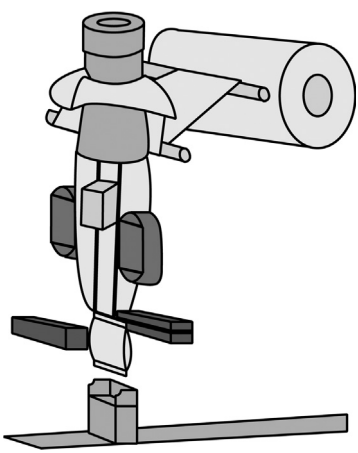


FIG. 13.11 Form fill seal machine

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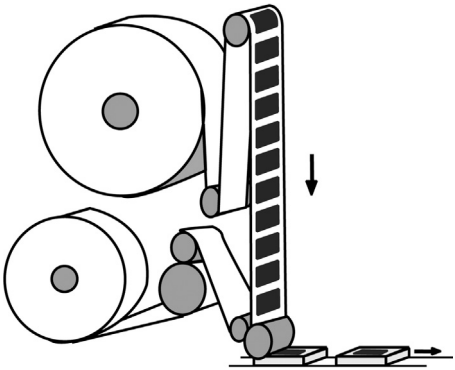


FIG. 13.12 Labeling a product

13.3. LABELING

Another broad area of packaging need is the area of labeling a product. Labels can be glued on, self-sticking (Fig. 13.12), or heat shrink wrapped (Fig. 13.13). Placing labels on flexible products like bags of potato chips is not easy, so much effort has been made to develop great printing processes onto the bag material itself.

13.4. CASES

Cases are made from corrugated cardboard, and like cartons, shipped in their knocked down condition (Fig. 13.14). Cases need to be erected (Fig. 13.15) by pushing the two opposite corners to form a rectangle, and the bottom flaps folded and either glued or taped. A filled case is then closed by folding the top flaps and sealing (Fig. 13.16). The case erecting machine can use tape or glue, depending on the machine's configuration, and the top flaps sealed by either method, depending on what type of box sealer one purchases. Depending on the product being shipped in a case, and perhaps even how far the case is to be shipped, some

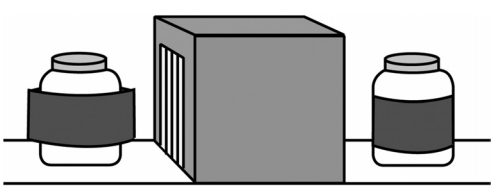


FIG. 13.13 Shrink fit labeling

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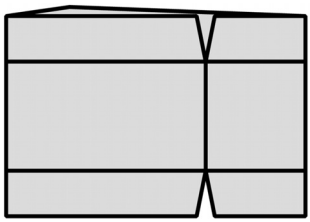


FIG. 13.14 Knocked down case

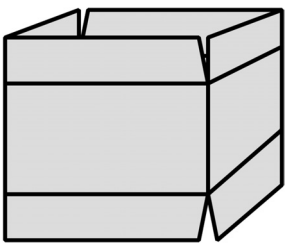


FIG. 13.15 Erected case

users have a personal preference for box sealing that is both logical and emotional. Cases are most often loaded from the top, but some applications use side loading of a single tier or layer in order to achieve the best packaging results (Fig. 13.17).

13.5. PALLETIZING

A pallet of cases can look something like Fig. 13.18. Here the cases are stacked in an alternating pattern on adjacent layers, and have a top layer or sheet to assist with lateral stability. Also in this figure, bands have been applied to lock the product into place. This method tends toward the extreme, and was more likely

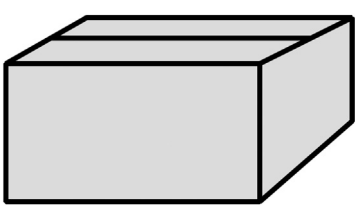


FIG. 13.16 Closed case

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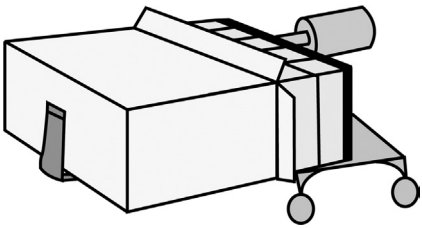


FIG. 13.17 Single tier loading

found in the past. Presently, plastic shrink wrap or stretch wrap is more often used. Wrapping can be done by either a machine or a human, and can be removed without the potential danger of cutting metal bands and watching them fly in all directions with their razor sharp edge that results from the cutting process.

Pallets traditionally are made of wood, but the longer life of molded plastic is becoming more accepted (Fig. 13.19). Wooden pallets are used more than once, but often do not last more than three transfers due to loading and forklift truck operator abuse. Cases on pallets can be placed in packing patterns based on the size of the pallet and the size of the cases. There are dozens of possible patterns, with some of the more basic ones shown in Fig. 13.20.

Pallets can be filled by the row (Fig. 13.21) or by the box (Fig. 13.22). There are all levels of complexity and speeds available depending on product

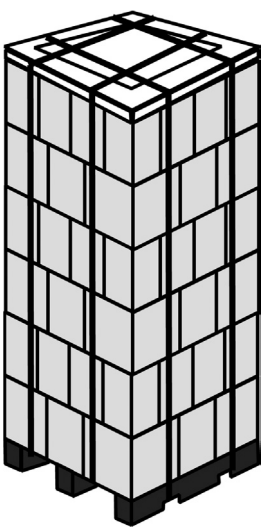


FIG. 13.18 Pallet of cases

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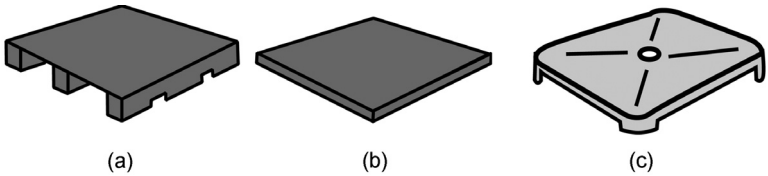


FIG. 13.19 Pallet forms: (a) wood; (b) solid slab; (c) molded plastic

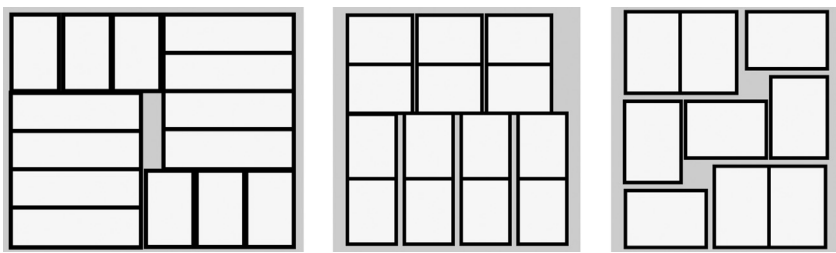


FIG. 13.20 Pallet packing patterns

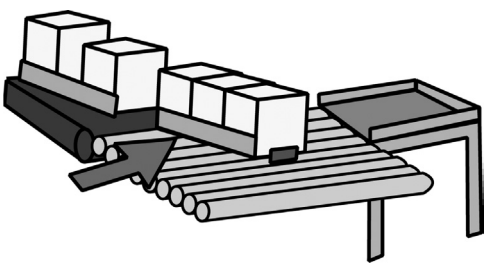


FIG. 13.21 Pusher bar moving a row of cases onto a pallet layer

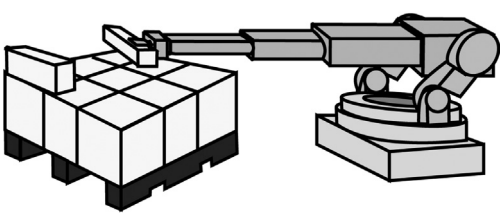


FIG. 13.22 Robotic palletizer

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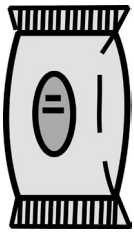


FIG. 13.23 Pouch

throughput. A cereal manufacturer making corn flakes will crank out many cases each hour, and can justify a high-speed palletizer that moves much faster than a robotic application.

13.6. FORMING POUCHES

There are many types of pouches in use today. Some of the better designs of yesteryear (Fig. 13.23) are now limited to being placed in a carton (Fig. 13.24). This older style pouch does not stand up by itself on a shelf, and does not grab the buyer's attention. Even fin seals (Fig. 13.25) and lap seals (Fig. 13.26) are old news.

Traditional pouch fillers (Fig. 13.27) have been around for decades, but the new Stand Up Pouch (SUP) has grabbed a lion's share of the market. Each year billions of SUPs are sold in the United States. Machines to fill SUPs have been developed, but the handling of both the empty SUP pouches and the filled SUPs does cause trouble.

13.7. BLISTER PACKS

Blister packs are not new to the consumer, but are still very much in demand and are likely to stay that way. A range of products can be packaged in blister packs

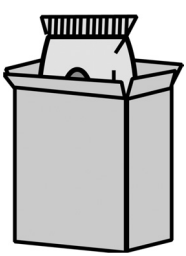


FIG. 13.24 Pouch in a carton

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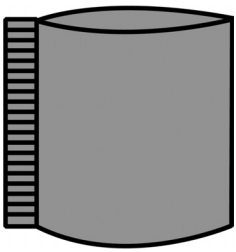


FIG. 13.25 Fin seal

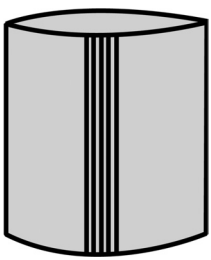


FIG. 13.26 Lap seal

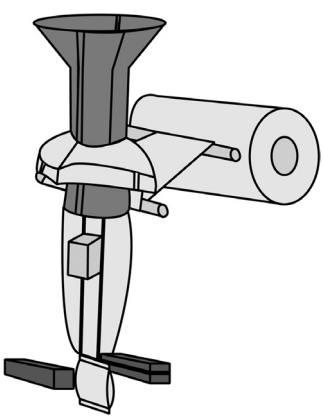


FIG. 13.27 Web feeding for pouch formation

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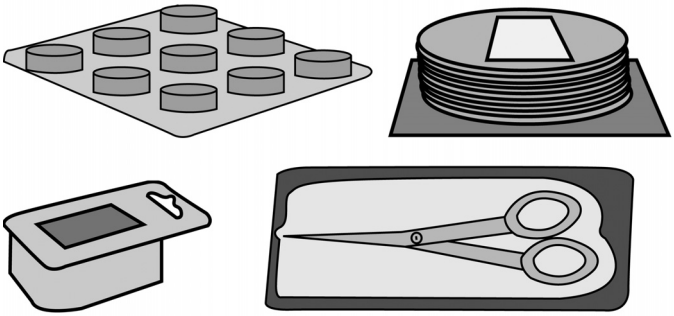


FIG. 13.28 Blister packs

(Fig. 13.28), some for internal use inside a carton (medicines and pills) while other products hang on racks using the blister pack as a way to show off the product without exposure to the elements.

The blister pack can be sealed to a card (Fig. 13.29) or it can be sealed with a foil, usually producing a more weather resistant package.

13.8. BAGS

Bags of one form or another have been around for centuries (Fig. 13.30). The paper bag is relatively new compared to the use of animal skins by cavemen. The filling of paper bags produces a product that is not very sexy compared to SUPs and modern packaging, but products like lawn fertilizer and bird food will continue to come in paper bags, at least in large quantities.

Most bags are filled by net weight, and use low-pressure high-volume air to blow the product into the bag's opening (Fig. 13.31). Some bags like lawn fertilizer blow into patented designed bags that actually self seal when they are filled.

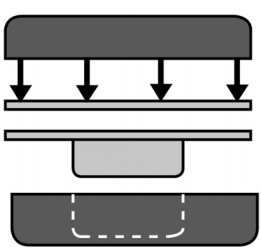


FIG. 13.29 Card feeding

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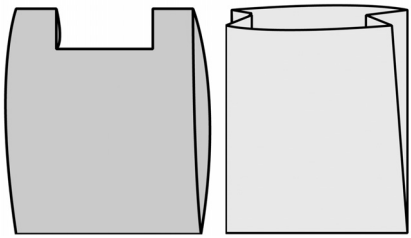


FIG. 13.30 Bags

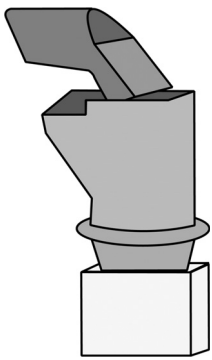


FIG. 13.31 Net weight filling

13.9. CONCLUSIONS

To many engineers, the package a product comes in is not very critical to one's purchasing decision. An engineer can usually think of what the product is, and because or despite the packaging, decide to purchase it, probably as a repeat customer, but since engineers are only 3–4% of the United States population, and are not a good representative of the purchasing public, packaging plays a huge role in how a product lives or dies.

Packaging continues to become more flashy, high tech, and sometimes even less recyclable, but unless the product beats out the competition and gains market share, unfortunately recycling is not always high on the list of concerns. So, the evolution of packaging has mixed results towards recycling, a travesty in the author's opinion. On the positive side, new packaging drives the automation market for new machines to be designed and built.

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